Indigo Newark

Cambridge / Nottingham | ijnewark@icloud.com | +44 7874 025867

linkedin.com/in/indigonewark · github.com/ijnewark · substack.com/@indigonewark

Research profile

MSc student in Computational Neuroscience, Cognition and AI, and First-Class Neuroscience graduate. I am interested in how biological and artificial learning systems represent and act on uncertainty: precision weighting and predictive processing in the brain, and uncertainty quantification and adaptive learning in machine learning. My work pairs mechanistic modelling (drift-diffusion, spiking and attractor models) with machine learning, and I am building towards research on uncertainty-aware learning in non-stationary environments and, more broadly, neuro-symbolic approaches to interpretable reasoning.

Education

MSc Computational Neuroscience, Cognition and AI | University of Nottingham | September 2026–present (expected completion 2027)

- Autumn study: Computational Cognitive Psychology and Neural Computation, covering connectionist and deep-network models of perception, language and memory, biophysical neuron models, Hopfield and attractor networks, neural coding and synaptic plasticity.
- Autumn quantitative training: supervised and unsupervised learning (regression, classification, density estimation) built on linear algebra, optimisation and probability; biomedical modelling with differential equations, phase-plane analysis, stochastic simulation and model fitting.
- Planned spring study: deep learning (CNNs, RNNs, generative models and LLMs); reinforcement learning (MDPs, Bellman equations, Q-learning and policy gradients); Biomedical Modelling in an AI World, including neural networks and brain dynamics.
- Research project: 60-credit supervised dissertation, summer 2027.

BSc Neuroscience, First-Class Honours | University of Nottingham | 2023–2026

- Final-year average 72.2% (credit-weighted).
- Current Research in Neuroscience 77%; predictive-coding perspective article 85%; comparative pain-processing minireview 85%.

Research experience

Computational proteomics dissertation | University of Nottingham | Supervisor: Dr G. Sheridan

- Built a computational pipeline to analyse hippocampal proteomics data, investigating how systemic infection relates to pathways implicated in Alzheimer's disease.
- Used Python, STRING and ProVision; wrote a reusable power-calculation tool for sample-size estimation (github.com/ijnewark/powercalc_dissertation).

Theoretical work

Predictive coding and autism | Assessed perspective article (85%), developed further on Substack

- Independently proposed a four-stage conceptual model of autism emergence using predictive coding and Bayesian precision weighting.
- Integrated developmental psychology and computational neuroscience literature into a theoretical account of atypical cognition, then developed it further on Substack (substack.com/@indigonewark).

Selected computational projects

Multimodal AI for neurological disease monitoring | Research proposal | 2026

- Designed an architecture combining wearable, imaging and behavioural data for longitudinal disease-state tracking, using contrastive pretraining, Bayesian change-point detection and conformal prediction for uncertainty quantification.
- Submitted to Bristol's Practice-Oriented AI CDT; reached the final-stage interview.

Decision making under conflict | Python | github.com/ijnewark/Decision-Making-Under-Conflict

- Implemented a four-parameter drift-diffusion model from scratch and fitted it to congruent and incongruent trials in the Hedge et al. (2018) flanker dataset (~200 participants), validating fits with parameter recovery. The congruency effect was carried by a lower drift rate on incongruent trials, with boundary separation similar across conditions.

Computational neuroscience portfolio | Python | github.com/ijnewark/computational_neuroscience_portfolio

- Implemented leaky integrate-and-fire neurons with F–I curve analysis, spike-timing-dependent plasticity, mutual and Fisher information, population decoding (PCA, logistic regression) and spike-train analysis across five modules, each with a script and output figures.

Technical skills

- **Programming and tools:** Python, R, MATLAB, Swift; HTML, CSS, JavaScript; PyTorch, TensorFlow, pandas, SciPy, Jupyter; GitHub, SQLite.
- **Methods:** neural and cognitive modelling, parameter estimation and system dynamics, dimensionality reduction, uncertainty quantification, statistical power and sample-size estimation.

Additional experience, outreach and awards

- **Private tutor**, Cambridge | September 2021–present: long-term academic support for a neurodivergent student, adapting materials and communication to their needs.
- **Freelance web developer**, Cambridge | April 2026–present: designed, built and deployed sendsage.co.uk, a nine-page responsive website (HTML, CSS, JavaScript) for an independent SEND advocacy service, with a contact form, structured-data markup and light/dark mode; iterated content and structure with the founder for families navigating the SEND system.
- **Stock Control and Pharmacy Assistant**, Tesco | October 2021–present: stock control, compliance checks, supervised pharmacy support and training new staff.
- **Independent instrument trading** | January 2025–present: Python-based market analysis for buying and resale decisions (github.com/ijnewark/reverb-analytics).
- **Neuroscience outreach**, University of Nottingham | February–March 2026: designed and delivered sensory-science activities for two Year 6 classes, adapting to different access needs.
- **Community volunteering:** redistributed surplus food to local charities, including Emmaus Cambridge.
- **Long Road Sixth Form College** | 2021–2023: A levels in Psychology A* (top of cohort), Chemistry A, Biology A; NHS Ethics Essay Award; Chemistry Research Project Award.
- **Wider interests:** writing on neuroscience, AI and philosophy of mind; music composition and production.